



Master's Thesis Defense

# ReMerkle: Proof of Reputation-driven Federated Learning with Compact Merkle Verification for Consumer-centric IoMT

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# Introduction & Context

## Global Aging Crisis

### WHO Statistics:

- By 2025: **1.2 billion** people aged 60+
- By 2050: **2 billion** people aged 60+

Chronic illnesses, physical impairments, and mental disorders are more prevalent in aging populations, increasing pressure on medical resources. [1]

## IoMT & Consumer Electronics

Wearable Body Area Networks (WBANs) connect tiny sensors (accelerometer, heart rate monitor) that capture vast amounts of **private physiological signals**.

Shift from **hospital-centric** to **home-centric** care models. [2]

## Federated Learning (FL)

Emerged as preferred solution:

- Introduced by Google in 2017 [3]
- Each device trains ML model **locally**
- Sends only **gradients/weights** to central server
- **Raw data never leaves** the device

## The Gas Cost Problem

Blockchain provides public, immutable record but faces critical limitations:

- Writing full update to Ethereum **costs more than consumer services are worth**
- Forcing every wearable to check each entry **drains tiny batteries**

# Related Works

This slide reviews **BCFL frameworks**, **reputation-based FL**, and **healthcare/IoMT BCFL** literature.

## 1 Blockchain-centric FL Frameworks

Surveys [4][5] classify BCFL architectures by auditability, decentralization, incentives, and scalability.

Representative systems include hierarchical IoV BCFL [6], LAFED [7], Ethereum-based OpenFL [8], and cloud-edge BCFL [9].

Consumer IoMT/CE work emphasizes adaptive, energy-aware, telemedicine FL [10][11][12] with incentive and security primitives [13][14].

## 2 Reputation-Based FL

Song et al. [15] introduced reputation-guided scheduling in secure wireless FL to improve resilience against malicious users.

Wang & Kantarci [16] extended reputation to mobile client selection and aggregation weighting.

Zhang et al. [17] applied reputation scoring to EEG classification; blockchain-backed reputation was also explored for MEC and O-RAN.

## 3 BCFL for Healthcare & IoMT

Privacy, provenance, and secure monitoring are addressed across smart healthcare, fraud-aware IoMT, real-time patient monitoring and wearable/fog-IoT platforms [17][18].

These efforts validate BCFL for healthcare workflows, but they do not yet target the lightweight, consumer-centric setting.

# Challenges & Research Gaps

## 1 Model Poisoning

Compromised device generates **crafted gradients** that degrade global performance or inject backdoors into the shared model.

## 2 Data Poisoning

**Corrupted local data** yields harmful updates from faulty or miscalibrated sensors, compromising model integrity.

## 3 FedAvg Limitations

Standard aggregation **treats all updates equally** — no mechanism to distinguish trustworthy contributors from malicious actors.

## 4 Sybil Attacks

Attackers register **multiple fake identities** to perform coordinated attacks, increasing chances of malicious updates being included.

## 5 Accountability Gap

Tracing which clients contributed what in decentralized systems is **nearly impossible**, yet regulatory compliance demands auditability.

## 6 Domain Specificity

Current reputation mechanisms treat all data-types the same via **generic indicators**; biomedical signals need specialized plausibility filters.

💡 **Research Gap:** No existing solution combines lightweight verification, fine-grained reputation filtering, and compact on-chain commitments for cost-sensitive consumer IoMT deployments.

# Research Goals & Contributions

**ReMerkle:** A reputation-based framework for consumer IoMT that treats blockchain as a "Digital Judge" rather than a data storage layer.

## 1 Custom IoMT Benchmark Dataset

Collected using **Raspberry Pi 4.0** consumer IoMT node. Implemented XGBoost, Random Forest, Voting Ensemble, and ANN for activity recognition.

## 3 Smart Contract-Driven Validation

Combines validation gain, cross-round consistency, biomedical plausibility with **reputation-weighted FedAvg** and client filtering.

## 5 Solidity Auditing Mechanism

Smart Contract uses **Merkle Paths** to validate data correctness. Verification cost **near-zero (<1ms)** with mathematical proofs against tampering.

## 2 Three-Tiered Reputation Gating

Chain of filtering stages: **Cosine Similarity** (Gate 1), **Beta Distribution History** (Gate 2), **Entropy-based Quality** (Gate 3).

## 4 Compact ReMerkle Architecture

Gas-saving storage: federates updates into Merkle Tree. Only **32-byte Merkle Root** saved on-chain achieving  **$O(1)$  storage complexity** per round.

# Dataset Overview

## IoMT Benchmark Dataset

Created to reflect **real-world, continuous physiological data** that IoMT nodes encounter in everyday life — not controlled laboratory conditions.

Total Samples

72,094

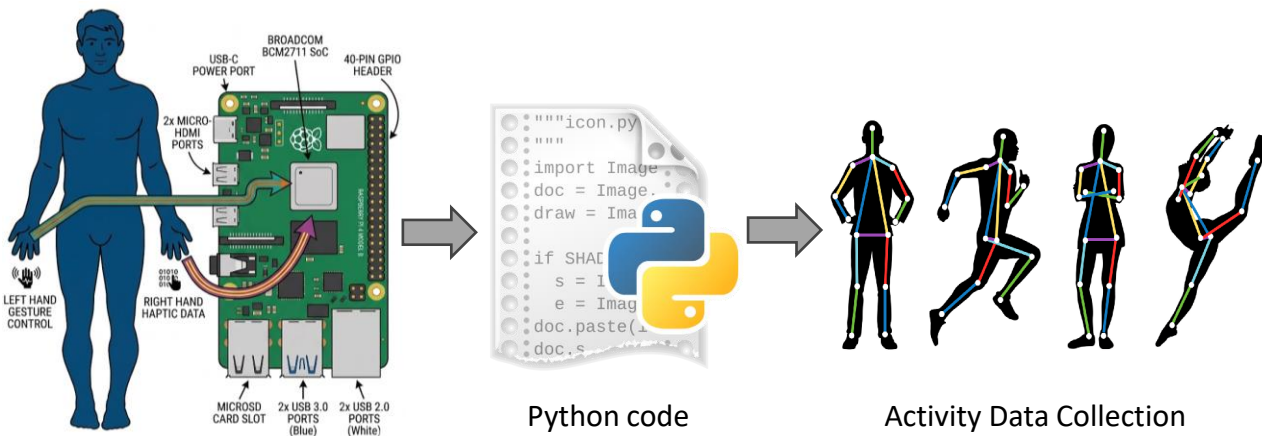
Sampling Rate

~50 Hz

Data Collection: Raspberry Pi 4.0 with accelerometer sensor

Participants: 6 persons

Split: 80-20 train-test with stratification

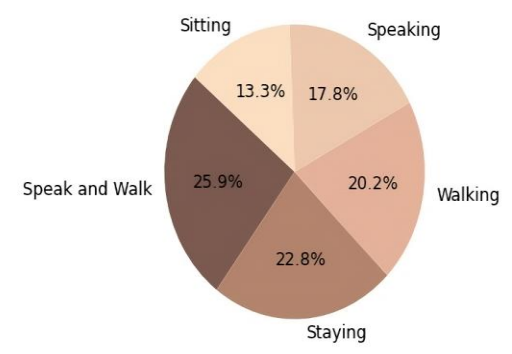
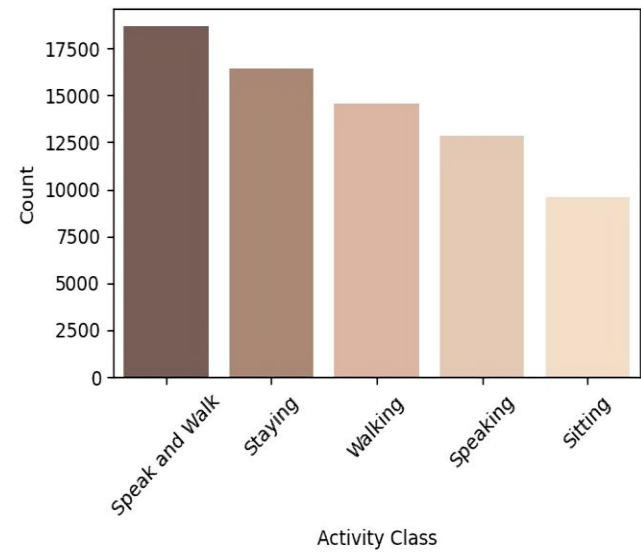
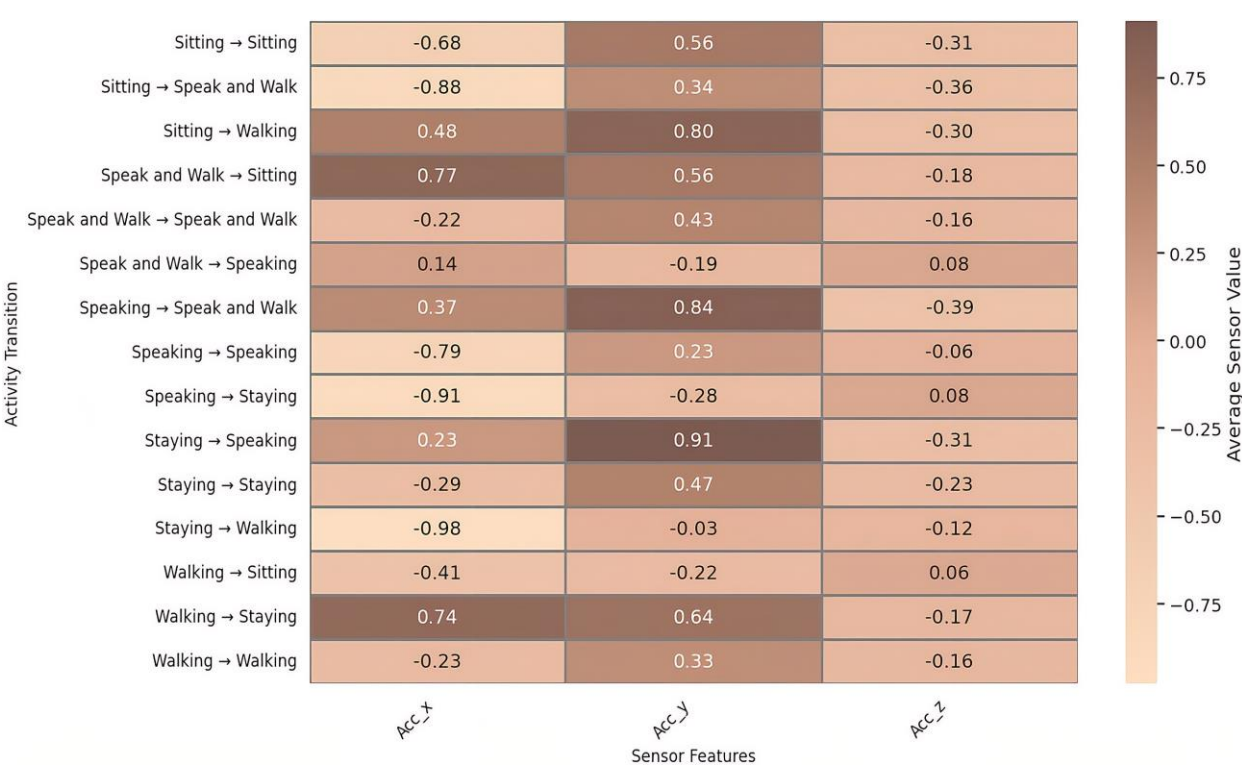


Python code

Activity Data Collection

## Five Activity Classes

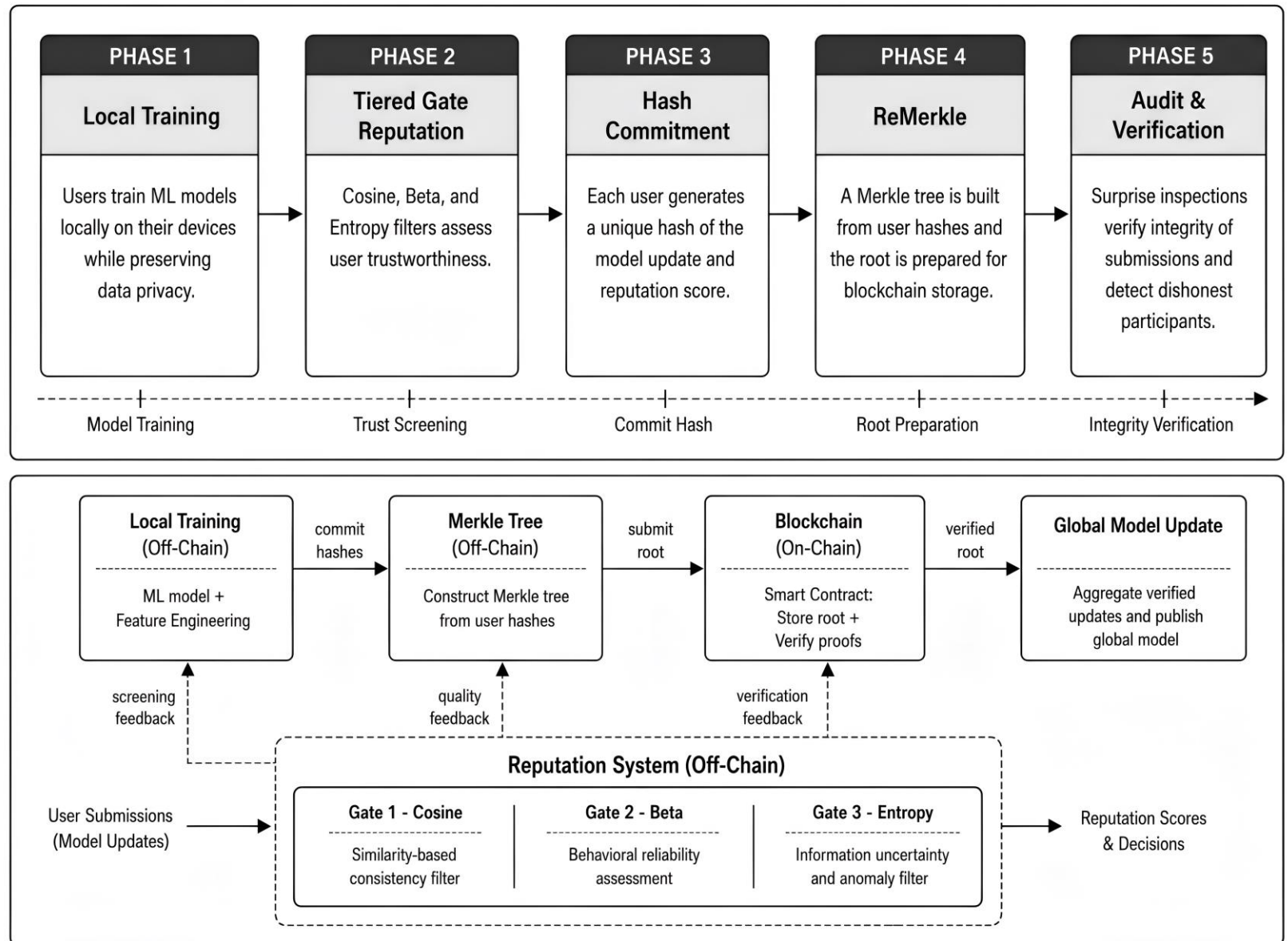
|                                                                                                  |                 |
|--------------------------------------------------------------------------------------------------|-----------------|
|  Speak & Walk | 18,674<br>25.9% |
|  Staying      | 16,432<br>22.8% |
|  Walking      | 14,561<br>20.2% |
|  Speaking     | 12,833<br>17.8% |
|  Sitting      | 9,594<br>13.3%  |



# Data Visualization



# System Overview: 5 Phase ReMerkle Pipeline



# System Overview: 5 Phase ReMerkle Pipeline (contd.)

## Key Innovation

### Blockchain as "Digital Judge"

Not a data storage layer — only reputation and audit metadata on-chain

### 99.57% Cost Reduction

From ~4.7M gas to ~20K gas per round

### $O(\log n)$ Verification

Merkle proofs enable efficient auditing

### < 2 seconds per round

Even with 100 participating nodes

## 1 Local Training

Each node trains locally on sensor data and creates **SHA-256 hash commitments** of local model updates.

## 2 3-Tiered Reputation Gating

**Gate 1:** Cosine Similarity check (threshold 0.3)

**Gate 2:** Beta Reputation check (threshold 0.2)

**Gate 3:** Entropy Quality check (penalty if  $H_i > 1.2$ )

Only accepted nodes proceed to next →

## 3 Merkle Tree Construction

Build Merkle tree from accepted commitments and extract **32-byte Merkle Root**.

## 4 Smart Contract Audit

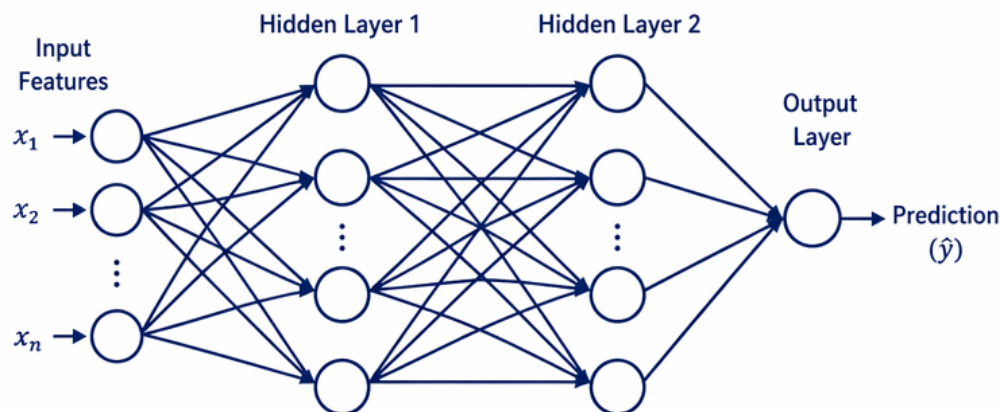
Store Merkle Root on blockchain. Random audit (20% of nodes) via Merkle Paths.

## 5 Global Aggregation

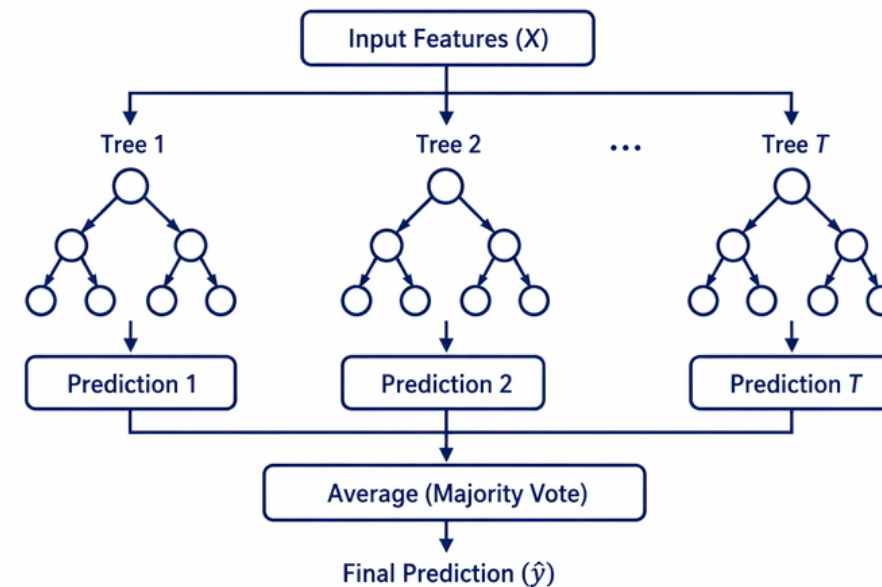
Reputation-weighted averaging of accepted updates to update global model.

# Local Model Training

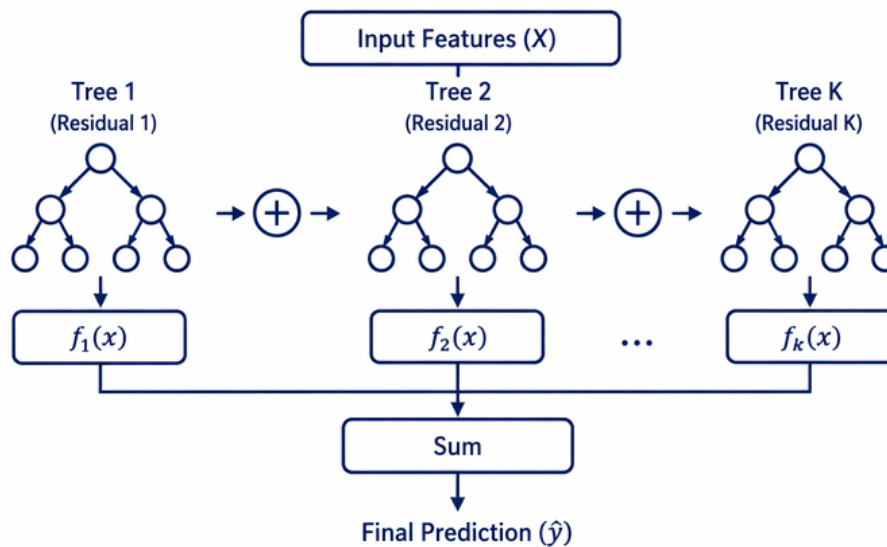
## 1. NEURAL NETWORK



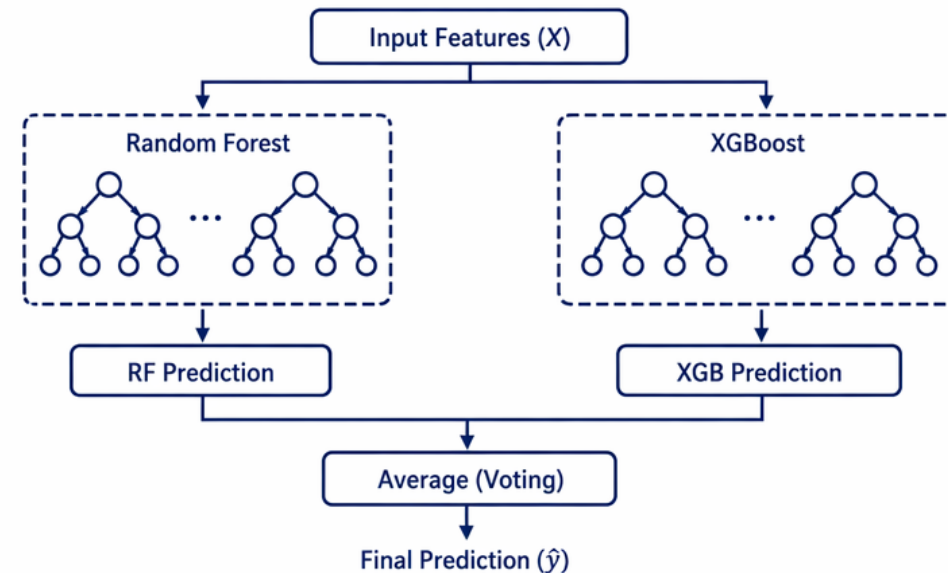
## 2. RANDOM FOREST



## 3. XGBOOST



## 4. VOTING ENSEMBLE (RANDOM FOREST + XGBOOST)



# Three-Tiered Reputation Gating

**1 Gate 1: Cosine Similarity**  
Fast alignment check — milliseconds detection, low cost

**2 Gate 2: Beta Reputation**  
Historical performance tracking over time

**3 Gate 3: Entropy Check**  
Prediction confidence evaluation

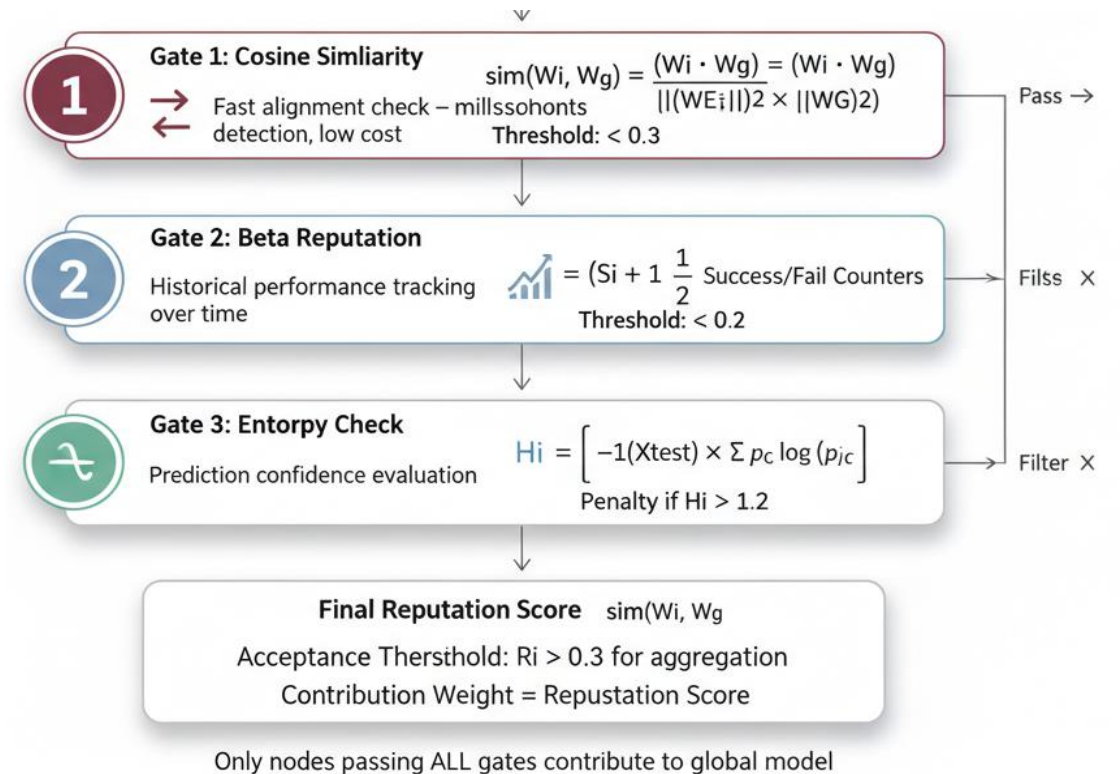
## Final Reputation Score

$$R_i = \beta_i \times \text{penalty}_i \times \text{sim}(W_i, W_g)$$

**Acceptance Threshold:**  $R_i > 0.3$  for aggregation

**Contribution Weight:** Equals reputation score

Example:  $R_i = 0.85$  has  $1.7\times$  influence vs  $R_i = 0.5$



## Gate Flow Logic

- Only nodes passing **all gates** contribute to global model
- Early rejection saves computation for obvious attacks
- Sequential design maintains strong security guarantees

# Gate 1: Cosine Similarity

## Purpose

Measures **alignment** between local update  $W_i$  and global model  $W_g$ . Detects model poisoning attacks by identifying gradient directions that deviate significantly from consensus.

## Formula

$$\text{sim}(W_i, W_g) = (W_i \cdot W_g) / (\|W_i\|_2 \times \|W_g\|_2)$$

## Vectorization for Random Forest

Tree parameters transformed into vectors:

```
Wi = flatten([T1params, T2params, ..., Tnparams])
```

## Threshold & Detection

### Threshold: 0.3

Nodes with score < 0.3 rejected immediately

- ✓ **Honest Nodes**  
Maintain reputation > 0.50 throughout training
- ✗ **Poisoned Updates**  
Typically have reputation < 0.2 (gradients opposing each other)

## Performance

- ⚡ **Detection time:** Within milliseconds
- ⚙️ **Computational cost:** Low (vector dot product)

# Gate 2: Beta Reputation

## Purpose

Records each node's **past performance history** using two counters per node. Derived from Beta distribution for modeling probability given success/failure counts.

## Beta Reputation Formula

$$\beta_i = (S_i + 1) / (S_i + F_i + 2)$$

New nodes start at 0.5 — no prior information bias

## Two Counters Per Node

### Success Count ( $S_i$ )

Incremented when node passes all gates and audits

### Failure Count ( $F_i$ )

Incremented when node fails any gate or audit

## Update Rules

✓ **Pass Gate 1 + Audit:**  $S_i = S_i + 1$

✗ **Fail Any Gate:**  $F_i = F_i + 1$

Acceptance Threshold:  $\beta_i \geq 0.2$

## Graceful Degradation

Single failure doesn't eliminate nodes immediately

Repeated failures compound exponentially

After **3 consecutive failures**, score plummets below threshold (0.17)

Node **permanently blocked** from future participation

📈 **Result:** Honest nodes' scores rise from 0.5 to 0.85 over 10 rounds

# Gate 3: Entropy Quality Check

## Purpose

Evaluates **prediction confidence** on node's local test set. Measures how certain the model is about its predictions — low confidence indicates potential data quality issues.

## Entropy Formula

$$H_i = -1/|X_{\text{test}}| \times \sum_j \sum_c p_{jc} \log(p_{jc})$$

$p_{jc}$  = predicted probability for class  $c$  of sample  $j$

## Interpretation

### High Entropy → Uncertain Model

Gives equal probabilities to multiple classes

### Low Entropy → Confident Model

Assigns most probability mass to one class

## Penalty Function

### If $H_i > 1.2$

High confusion / uncertain

0.6

### Otherwise

Confident predictions

1.0

## Threshold Selection

Value 1.2 selected empirically

For 5 classes, max entropy =  $\log_2(5) \approx 2.32$

1.2 corresponds to **moderate confusion**

**i** Nodes with entropy  $> 1.2$  get penalized with 0.6 weight, lowering their contribution to global model.

# Merkle Tree Architecture

## Problem

Traditional methods store **every node's full model** on-chain  
= Prohibitively expensive

## Solution

Compress **thousands of model updates** into single **32-byte hash**

## Merkle Tree Structure

**Leaf nodes:** Hash of transaction data

$$H_i = \text{SHA256}(\text{serialize}(W_i) \parallel H_i \parallel \text{entropy}_i)$$

**Non-leaf nodes:** Hash of children

$$H_{\text{parent}} = \text{SHA256}(\text{sort}(H_{\text{left}} \parallel H_{\text{right}}))$$

**Root node:** Single hash representing all data

## 3-Step Process

### Step 1:

Each node generates cryptographic fingerprint (hash commitment)

### Step 2:

Build Merkle tree from leaf hashes (pairwise hashing until root)

### Step 3:

Extract 32-byte Merkle Root and store on blockchain

## Cost Reduction

Traditional

**75KB**

5 nodes × 15KB

ReMerkle

**32B**

Merkle Root

**2,350×**

Cost Reduction



# Smart Contract & Gas Cost Analysis

## Smart Contract Verification

Only **Merkle Root** saved on-chain. Audited nodes provide:

- Original commitment  $H_i$
- Merkle Path as cryptographic proof

### Verification Algorithm:

Hash leaf with path elements → check equality with stored root

## Gas Cost Comparison

| Approach    | Gas Cost | Complexity  |
|-------------|----------|-------------|
| Traditional | ~4.7M    | $O(n)$      |
| Hash-Only   | $O(1)$   | No proof    |
| ReMerkle ✓  | ~20K     | $O(\log n)$ |

## Annual Cost Analysis

At 30 gwei, ETH=\$3,000 (3 rounds/day)

|                                      |           |
|--------------------------------------|-----------|
| Traditional                          | \$154,889 |
| \$141.45/round $\times 3 \times 365$ |           |
| ReMerkle                             | \$657     |
| \$0.60/round $\times 3 \times 365$   |           |

## Scalability

At 100 nodes:

| Cost Savings |          |            |
|--------------|----------|------------|
| Traditional  | ReMerkle | Difference |
| 100M gas     | 26K gas  | 5,000×     |

# Experimental Setup

## Hardware Configuration

### Client Device

Raspberry Pi 4.0  
Consumer IoMT gateway node  
Accelerometer sensor for data collection

### Server

CPU: Intel 13th Gen Core i9  
RAM: 128 GB  
GPU: NVIDIA RTX 5070 (12 GB VRAM)  
Storage: 1 TB NVMe M.2 SSD

## Software Stack

Python 3.13   TensorFlow 2.18   Scikit-learn 1.6   Jupyter Notebook

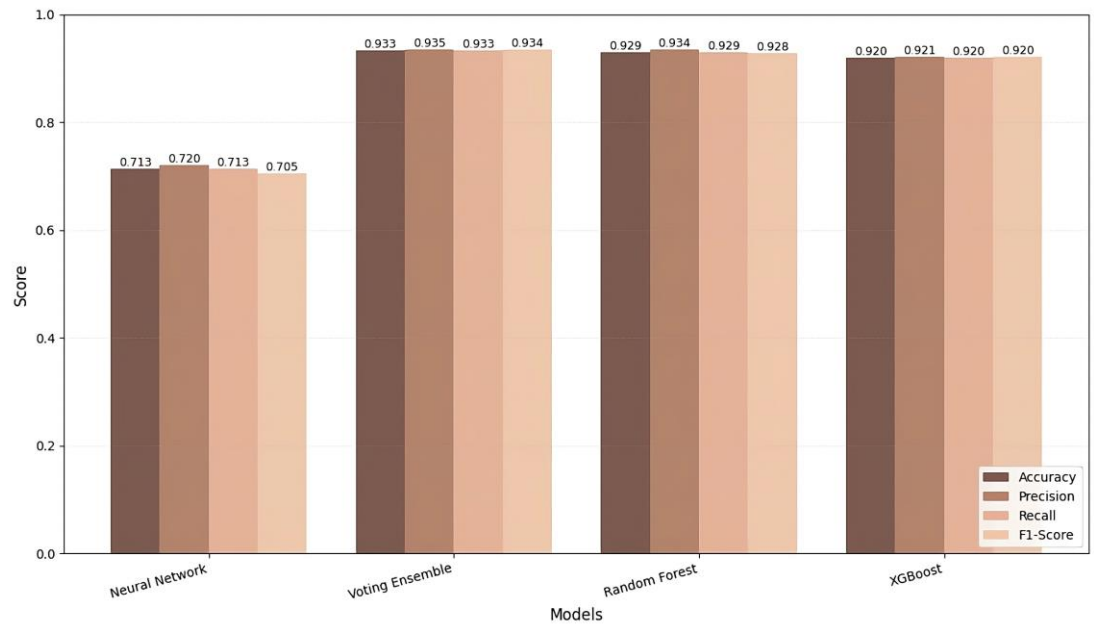
## FL Configuration

| Hyperparameter                   | Value                        |
|----------------------------------|------------------------------|
| Total Clients (N)                | 5                            |
| Communication Rounds (T)         | 30                           |
| Local Epochs (E)                 | 5                            |
| Batch Size (B)                   | 32                           |
| Optimizer                        | Adam                         |
| Learning Rate ( $\eta$ )         | $1 \times 10^{-3}$           |
| DP Noise Multiplier ( $\sigma$ ) | 0.01                         |
| DP Clipping Norm (C)             | 1.0                          |
| Client Sampling                  | Reputation-Proportional      |
| Aggregation Rule                 | Reputation-Weighted FedAvg   |
| DP Mechanism                     | Gradient Clipping + Gaussian |
| Loss Function                    | Categorical Cross-Entropy    |

# Results: ML Performance with Federated Learning

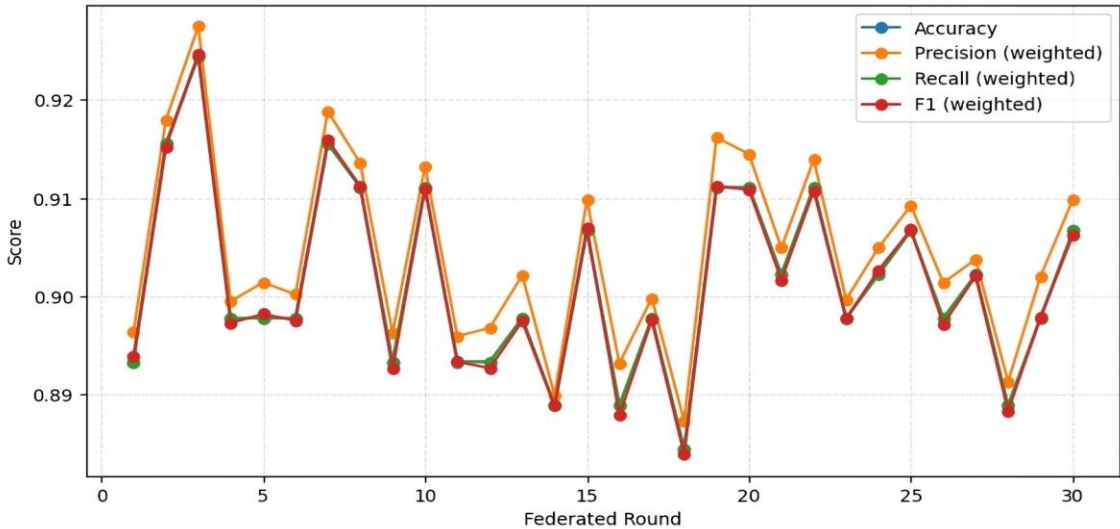
## ML Model Comparison

| Model             | Accuracy | Precision | Recall | F1    |
|-------------------|----------|-----------|--------|-------|
| Neural Network    | 0.713    | 0.720     | 0.713  | 0.705 |
| Random Forest     | 0.929    | 0.934     | 0.929  | 0.928 |
| XGBoost           | 0.920    | 0.921     | 0.920  | 0.920 |
| Voting Ensemble ✓ | 0.933    | 0.935     | 0.933  | 0.934 |



## 30-Round FL Accuracy

|          |       |
|----------|-------|
| Round 1  | 89.4% |
| Round 5  | 89.8% |
| Round 10 | 91.1% |
| Round 15 | 90.7% |
| Round 20 | 91.1% |
| Round 30 | 91.2% |



# Results: Blockchain Efficiency Gains

## Core Efficiency Result

Traditional Storage  
per round

1M gas

ReMerkle Method  
per round

20,139 gas

49.7×

Reduction (log scale)

## Cost Conversion

At 30 gwei, ETH=\$3,000:

Traditional

\$141.45/round

ReMerkle

\$0.60/round

## On-Chain Storage Growth

Traditional

Linear growth: 600KB by round 10

Hash-Only

1.5KB storing commitment hashes

ReMerkle

0.32KB total — only latest root (old roots overwritten)

99.96%

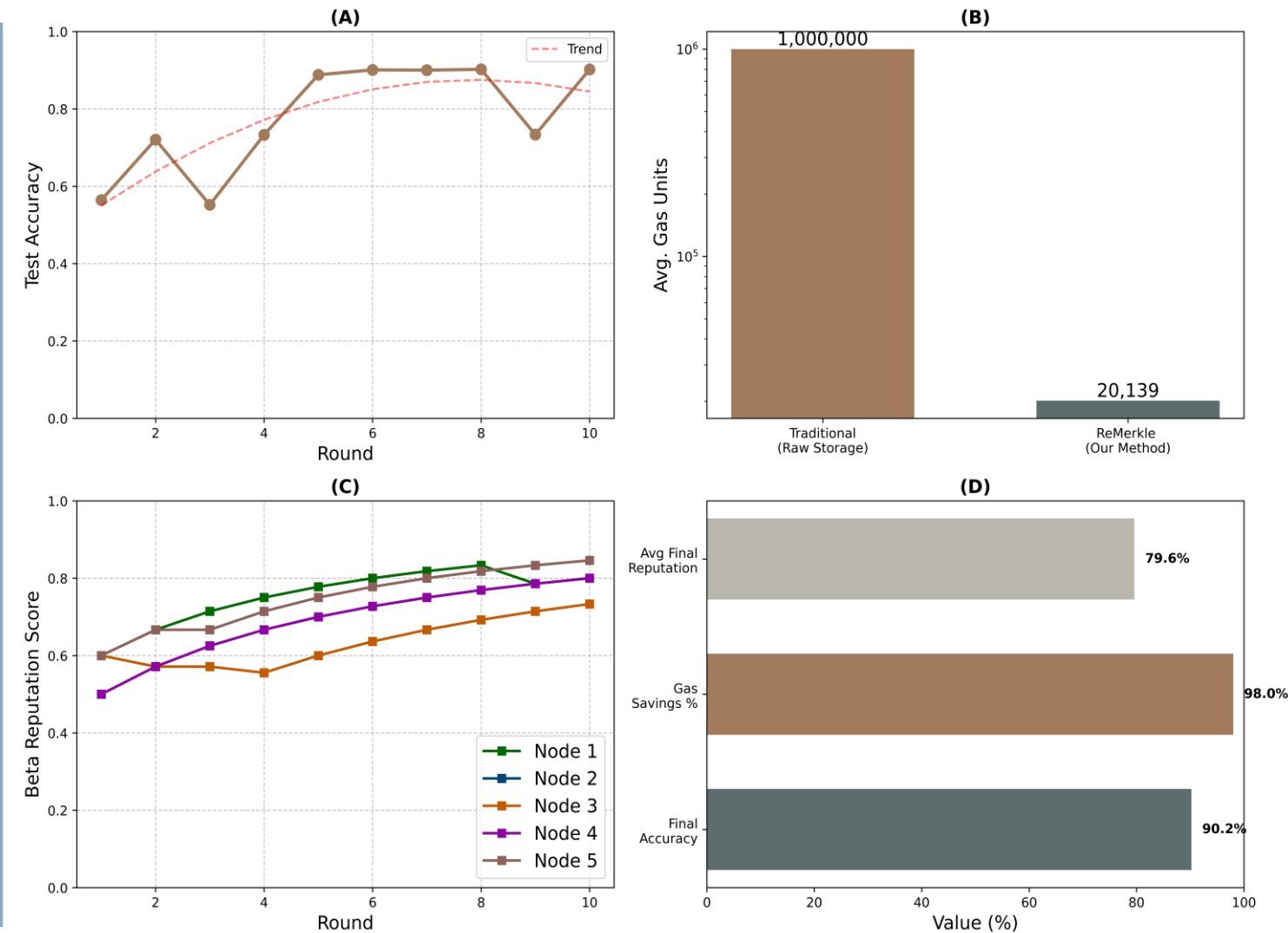
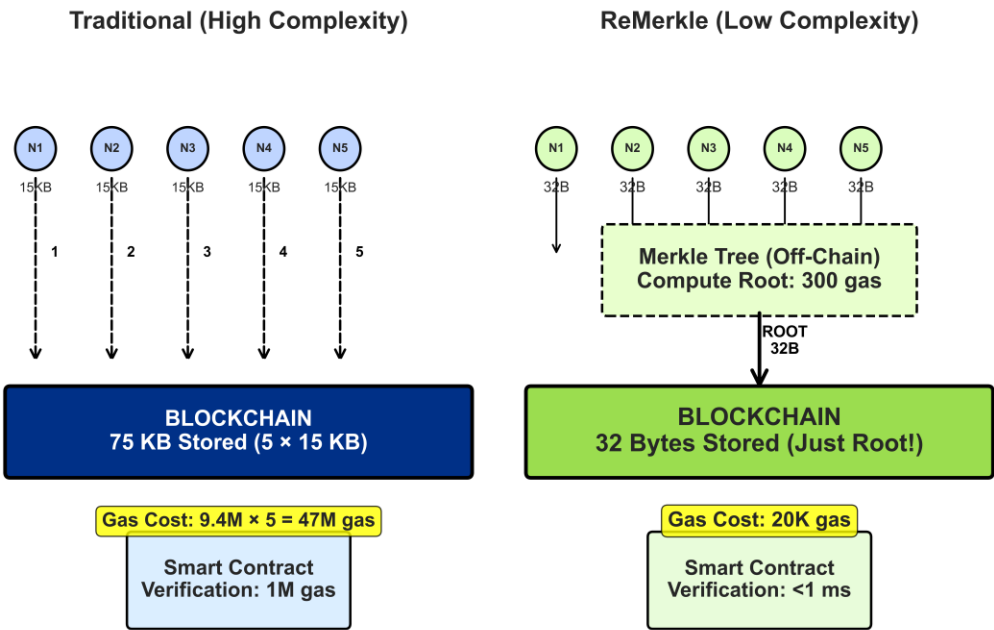
Storage Reduction

## Scalability (5-100 nodes)

Traditional: Linear  $O(n)$ , reaches  $10^7$  gas at 100 nodes

ReMerkle:  $O(\log n)$ , ~20,000 gas (+420 gas for tree height)

# Blockchain Efficiency Gains



# Security-Efficiency Tradeoff Analysis

## Quadrant Analysis

### Traditional (1, 9)

High security, low efficiency — unsustainable

### Hash-Only (9, 3)

High efficiency, poor security — no verification

### Optimistic Rollup (7, 7)

Moderate scores, delayed security adds latency

### ZK-Rollup (9.5, 8.5)

Excellent security, moderate efficiency cost

### Centralized (10, 1)

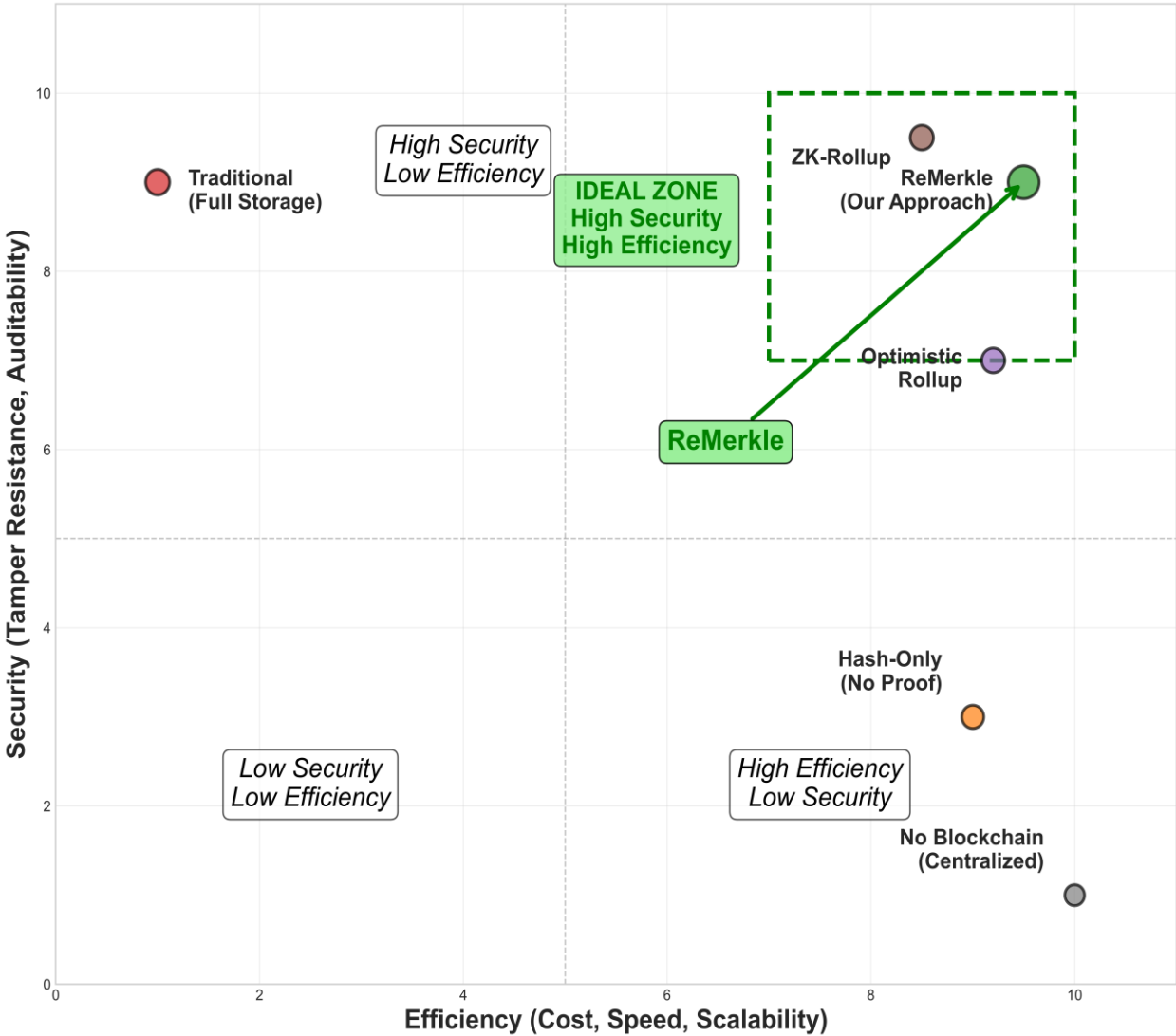
Maximum efficiency, minimal security

### ReMerkle: The Ideal Zone

Security Score 9.5

Efficiency Score 9.5

- ✓ High Security: Cryptographic proof verification
- ✓ High Efficiency: Logarithmic scaling



# Ablation Study: Robustness Against Label Noise

## Experimental Setup

5 federated clients with varying label noise levels to simulate real-world data quality degradation:

- C1

0% noise (Benign)

High-quality data source
- C2

10% noise (Minor)

Occasional sensor errors
- C3

30% noise (Moderate)

Significant quality issues
- C4

50% noise (Malicious)

Malicious adversary
- C5

70% noise (Byzantine)

Severe adversary

## Results

| Client   | Noise | Final Rep. | Status   |
|----------|-------|------------|----------|
| Client 1 | 0%    | 0.554      | Active   |
| Client 2 | 10%   | 0.290      | Filtered |
| Client 3 | 30%   | 0.116      | Filtered |
| Client 4 | 50%   | 0.131      | Filtered |
| Client 5 | 70%   | 0.141      | Filtered |

## Correlation Analysis

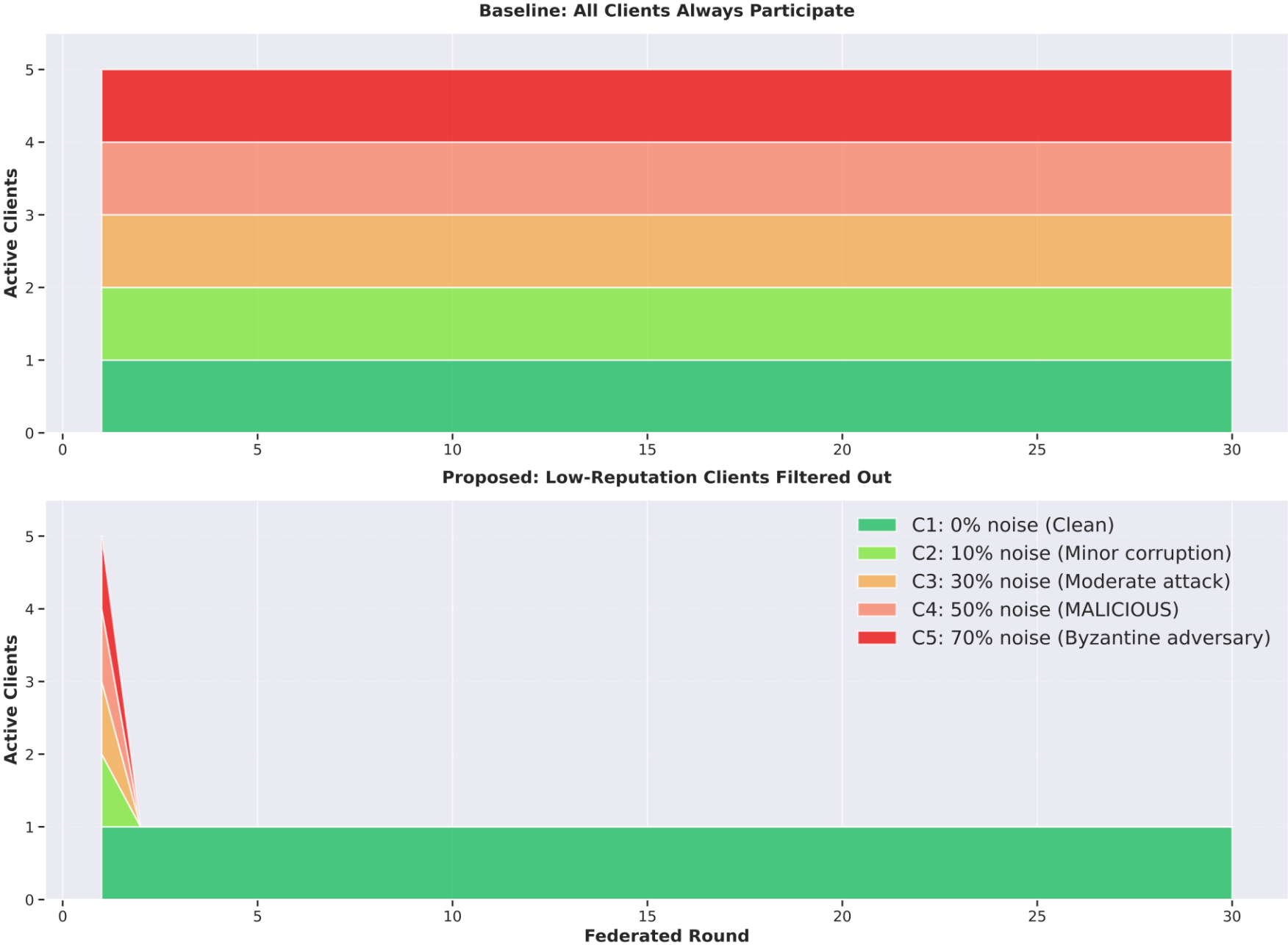
$r \approx -0.87$

Pearson correlation

$\rho \approx -0.87$  (Spearman)

Strong negative linear relationship — reputation scores monotonically decline as noise increases

Ablation Study:  
cont.





# Generalizability Test: ECG Arrhythmia Dataset

## Dataset Overview

**Source:** ECG Arrhythmia Classification Dataset (Kaggle)

**Samples:** 175,729 heartbeat recordings

**Features:** 32 ECG-derived features

**Feature Categories:**

- RR intervals (pre-RR, post-RR)
- Peak amplitudes (pPeak, tPeak, rPeak)
- Temporal intervals (QRS, PQ, QT, ST)
- QRS morphology descriptors

**Split:** 80/20 train-test, 5 federated clients (~28,117 samples each)

## Key Results

**Client 1 (0% noise)**

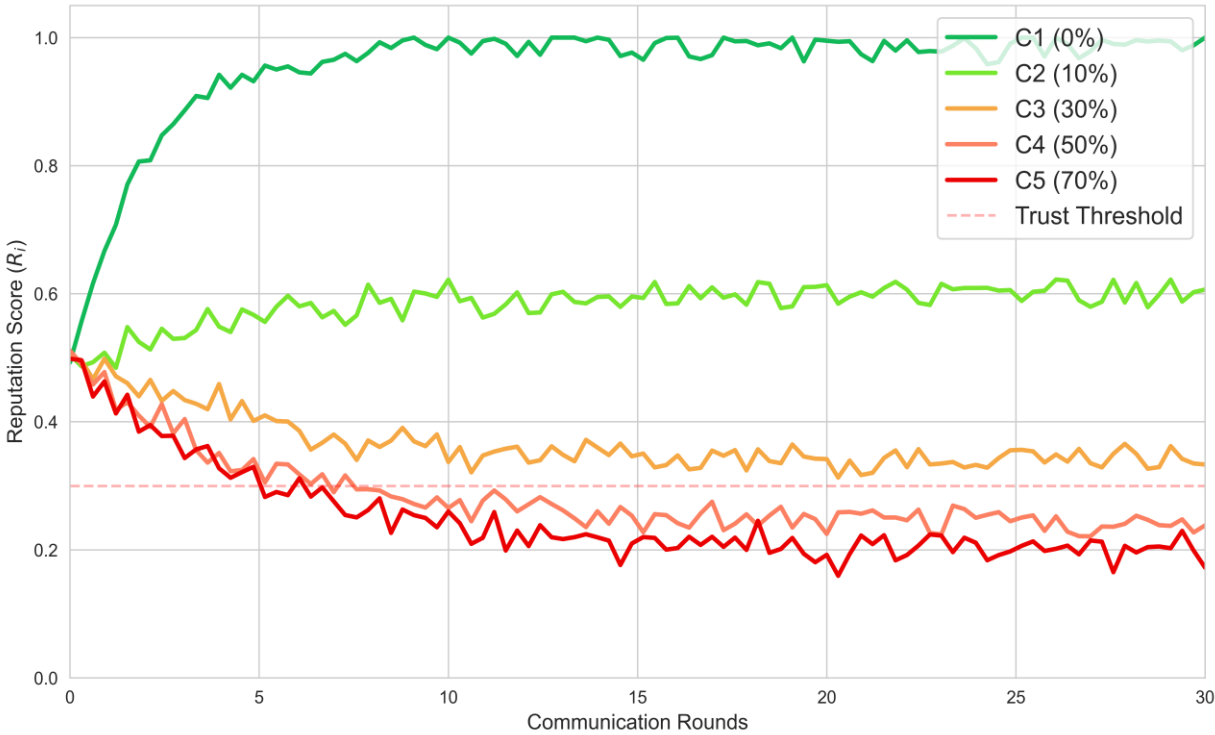
Near-perfect reputation: ~0.99

**Client 5 (70% noise)**

Drops below 0.3 threshold by Round 1

**Dynamic Filtering**

Two most corrupted clients excluded immediately (40% filtering rate)



## Noise Injection Pattern

Same pattern as IoMT experiment:

C1: 0%

C2: 10%

C3: 30%

C4: 50%

C5: 70%

# Comparison with state-of-the-art

| Reference                          | Domain             | Rep. Type                              | Poison.<br>Defence | Privacy            | Storage                | WBAN Focus | Incentive |
|------------------------------------|--------------------|----------------------------------------|--------------------|--------------------|------------------------|------------|-----------|
| Qu et al. [4, 5] (2022)            | General FL         | None                                   | ✗                  | Various            | High                   | ✗          | ✗         |
| Javed et al. [6] (2025)            | IoT / BCFL         | None                                   | Partial            | Lightweight verify | High                   | ✗          | ✗         |
| Almogadwy & Alqarafi [7] (2025)    | IoMT / HC 5.0      | None                                   | ✗                  | RTS-DELM privacy   | Med.                   | P          | ✗         |
| Bhasker et al. [8] (2025)          | IoMT / HC          | None                                   | P (IDS monitor)    | None stated        | High                   | ✗          | ✗         |
| Murthy & Shri [9] (2026)           | IoMT               | None                                   | ✓ (Byzantine)      | Convolutional AE   | Med.                   | ✗          | ✗         |
| Khan et al. [10] (2024)            | IoMT / HC          | None                                   | P (ensemble)       | Privacy-preserving | N/A                    | ✗          | ✗         |
| Wang et al. [11] (2025)            | CE                 | None                                   | ✗                  | Knowledge distill. | N/A                    | ✗          | ✗         |
| Wen et al. [12] (2025)             | CE (resource-ltd.) | None                                   | ✓ (Byzantine)      | Lightweight DP     | N/A                    | P          | ✗         |
| Shu et al. [13] (2026)             | IoT                | None                                   | ✓ (Byzantine)      | Joint DP + robust. | N/A                    | ✗          | ✗         |
| Song et al. [14] (2022)            | Wireless / IoT     | Basic (scheduling)                     | ✓                  | None               | N/A                    | ✗          | ✗         |
| Wang & Kantarci [15, 16] (2020/21) | Mobile / IoT       | Basic (selection)                      | P                  | None               | N/A                    | ✗          | ✗         |
| Cai et al. [17] (2025)             | General FL         | Multi-dim (gradient align., stability) | ✓ (strategic)      | None               | N/A                    | ✗          | ✓         |
| Rashid et al. [18] (2025)          | General FL         | Multi-dim (fairness)                   | ✓                  | None stated        | High                   | ✗          | ✓         |
| Chen et al. [19] (2024)            | IoT (async)        | Basic                                  | P                  | None               | Med.                   | ✗          | ✗         |
| Moyeen et al. [20] (2025)          | CE                 | Basic (hierarchical)                   | P                  | None               | N/A                    | ✗          | ✗         |
| Han et al. [21] (2025)             | IIoT               | Incentive-based                        | P                  | None               | High                   | ✗          | ✓         |
| Lu et al. [22] (2024)              | HC marketplace     | Rep.-incentive hier.                   | P                  | None stated        | High                   | ✗          | ✓         |
| Cai et al. [23] (2025)             | General FL         | Multi-dim                              | ✓                  | Dual DP + encrypt. | High                   | ✗          | ✗         |
| Deng et al. [24] (2025)            | General FL         | None                                   | P (trimming)       | None               | N/A                    | ✗          | ✗         |
| Hosain & Çakmak [25] (2025)        | IoMT               | None                                   | ✓ (XGB IDS)        | SHAP / LIME XAI    | N/A                    | P          | ✗         |
| Tawfik et al. [26] (2025)          | IoMT               | None                                   | ✓ (few-shot)       | DP (comm. —75%)    | N/A                    | ✗          | ✗         |
| Albogami [27] (2025)               | IoT Edge           | None                                   | P (DBN + op-       | Feature selection  | N/A                    | ✗          | ✗         |
| ReMerkle (Proposed)                | WBAN / IoMT        | 3-Tier Reputation                      | ✓                  | Rep. Merkle audit  | Gating + Const. (32 B) | ✓          | ✗         |

# Limitations & Future Work

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## Current Limitations

- 1 Simulated Blockchain**  
Real Ethereum mainnet may have additional gas costs from contract execution overhead
- 2 Fixed Audit Rate**  
20% rate balances cost/security for honest-majority; 30%+ malicious nodes may need higher rates
- 3 50Hz Sampling Assumption**  
Modern wearables support 100Hz+ requiring window size optimization
- 4 Estimated Costs**  
Preliminary estimates within 10% of simulated values but need pilot verification

## Future Directions

- 1 Adaptive Audit Strategies**  
Dynamically adjust verification rates based on detected threat levels using ML prediction
- 2 Cross-Device FL**  
Enable collaboration between heterogeneous wearables (Apple Watch, Fitbit, Android)
- 3 Layer 2 Integration**  
Optimism or Arbitrum for further cost reduction while maintaining security
- 4 Pilot Smart Contract Deployment**  
Measure actual on-chain costs and validate estimates

# Conclusion

## ReMerkle Framework

A federated learning framework with reputation-based blockchain technology **tailor-made for consumer IoT devices**. Blockchain acts as a **"Digital Judge"** not a data storage layer.

## Key Achievements

- ✓ **Blockchain Cost Reduction**  
Via Compact Merkle Tree Verification
- ✓ **Three-Tiered Reputation Gate System**  
Cosine, Beta, Entropy — scaffolds security while maintaining productivity
- ✓ **Storage Reduction**  
75KB → 32 bytes per round
- ✓ **93.3% Classification Accuracy**  
Over 5 activity classes

## Performance Highlights

### Gate 1: Cosine Similarity (Detection)

Model poisoning detected in **milliseconds**

### Gate 2: Beta (Tracking)

Consistent performance monitoring over time

### Gate 3: Entropy (Isolation)

Faulty outputs filtered by entropy analysis

### Speed

**< 2 seconds per round** even with 100 nodes

## Reputation Convergence

**0.5 → 0.85**

Honest node reputation after 10 rounds

# Publications & Resources

## IEEE Transactions on Consumer Electronics (Q1-Tier, IF-10.9)

### Paper 1 (Published)

"Towards Trustworthy Consumer-Centric IoMT: Reputation-Driven Federated Learning for Secure Medical Data Sharing." DOI: 10.1109/TCE.2026.3685994

M. I. Mahmud, J. Bieniek, M. Rahouti, A. Chehri, G. Jeon

### Paper 2 (Under Review)

"ReMerkle: Proof of Reputation-Driven Federated Learning with Compact Merkle Verification for Consumer-centric IoMT"

M. I. Mahmud \*, J. Bieniek \*, M. Rahouti, A. Chehri, G. Jeon

\* Equal contribution

## GitHub Repository

[www.github.com/Monirules/ReMerkle](https://www.github.com/Monirules/ReMerkle)

## Dataset

### Custom IoMT Benchmark Dataset

72,094 samples collected via Raspberry Pi 4.0

🔒 Available upon reasonable request with research proposal

## Contact

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# Thank You

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